21-02-2026 Initial Understanding

When I first read the problem statement, I understood that the system must:

- Accept multiple alternatives
- Accept multiple criteria
- Support weighted importance
- Generate a ranked recommendation
- Clearly explain the reasoning

The most important requirement was that the system should not be a black-box AI solution. It must be explainable and transparent.So this problem belongs to **Multi-Criteria Decision Making (MCDM) a**s per my deep research.

Research Phase

Before coding, I studied three approaches:

1. **Weighted Sum Model (WSM)**
2. **TOPSIS**
3. **AHP**

22-02-2026 Weighted Sum Model

It is like a normal weight bias algorithm in ML.
Score = sum of (weight × value)

It is easy to implement , but does not ,

- Requires manually assigned weights
- No consistency validation
- Too simplistic for complex decisions

So for basic initializatI started coded with this.

23-02-2026 TOPSIS

TOPSIS ranks alternatives based on distance from:

- Ideal best solution
- Ideal worst solution

It is mathematically strong and widely used. But,

- Weights must still be predefined
- No validation of judgment consistency

- Less focused on modeling human reasoning

Since the assignment emphasizes transparency and reasoning, I wanted something more structured.

24-02-2026 Final Choice – AHP

I selected **AHP (Analytic Hierarchy Process)** because,

- It derives weights using pairwise comparisons
- It models human reasoning
- It includes a built-in Consistency Ratio(I will research about this more in future for explainability)
- It makes the reasoning process visible

Unlike TOPSIS, AHP does not assume weights are known, it calculates them logically.

Also, I choose it because I want to do different unlike other people, I thought of Hybrid like Algorithm in future to implement in it.

Implementation Evolution

Version 1 – Simple AHP

- CLI input
- Pairwise matrix
- Weight calculation
- Basic ranking

Implemented the core logic.

Also studied Git to be implemented in terminal, all its procedure like git init, add, pull, push ,commit etc

25-02-2026 Improved Version

- Saaty scale guidance
- Input validation
- Consistency Ratio calculation
- Weight breakdown display
- Clear explanation of results

26-02-2026 After implementing the improved AHP version, I realized that real-world decisions are rarely static.

In practical situation,

- Users may slightly change their priorities
- Criteria importance may shift
- External constraints may evolve

A small change in weights can sometimes change the final ranking.

To test the robustness of the recommendation, I implemented a **Sensitivity Analysis version**.

How to implement it

1. Slightly increase or decrease each criterion weight (e.g., ±10%).
2. Recalculate final scores.
3. Compare the new ranking with the original ranking.
4. Highlight if the top alternative changes.

This allows us to,

- Identify dominant criteria
- Detect unstable decisions
- Provide deeper insight beyond ranking

While executing a test case on Job Selection, I found that Package and Layoff have complementary effect
After research on this topic, I understood about ,

Benefit criteria and **Cost criteria**.

- **Benefit (Positive Effect)** → Higher value = Better
  Example: Salary package, Growth, Learning opportunities
- **Cost (Negative Effect)** → Higher value = Worse
  Example: Layoff risk, Work pressure, Cost, Distance

So implemented it.

Handling both benefit and cost criteria ensures the system is applicable to real-world decision problems where trade-offs include both gains and risks.

27-02-2026

Today, I tested my results with Job Selection, then I understood complexity of my project of using a matrix, and understood problems for a user.

For example, in a job selection decision:

- Package (LPA) is an objective criterion.
  Since exact numeric values are available (e.g., 12 LPA, 10 LPA, 8 LPA), there is no need for pairwise comparison.
  The system directly normalizes these values and assigns weights automatically.
- Work Culture or Growth Opportunity are subjective criteria.
  Since they cannot be measured numerically, the system uses AHP pairwise comparisons to capture human judgment.

Hybrid Improvement – Objective + Subjective Criteria Handling

In the improved version of the system, I introduced a hybrid approach to handle two different types of criteria: objective and subjective.

This improvement reduces unnecessary comparisons for measurable factors while still preserving structured reasoning for subjective factors.

As a result, the system becomes:

- More efficient
- More realistic
- Less user-burdensome
- Better aligned with real-world decision-making

This hybrid design makes the Decision Companion both mathematically sound and practically usable.

28-02-2026

Job Selection	Stock Trading
Salary known	Returns uncertain
Layoff risk somewhat known	Risk highly dynamic
Culture subjective	Market sentiment unpredictable
Mostly stable	Highly volatile

So classical AHP is not fully enough.

Mean–Variance Model (Markowitz Portfolio Theory)

This is the foundation of modern portfolio theory.

Used when:

- Comparing multiple assets
- Considering return vs volatility trade-off

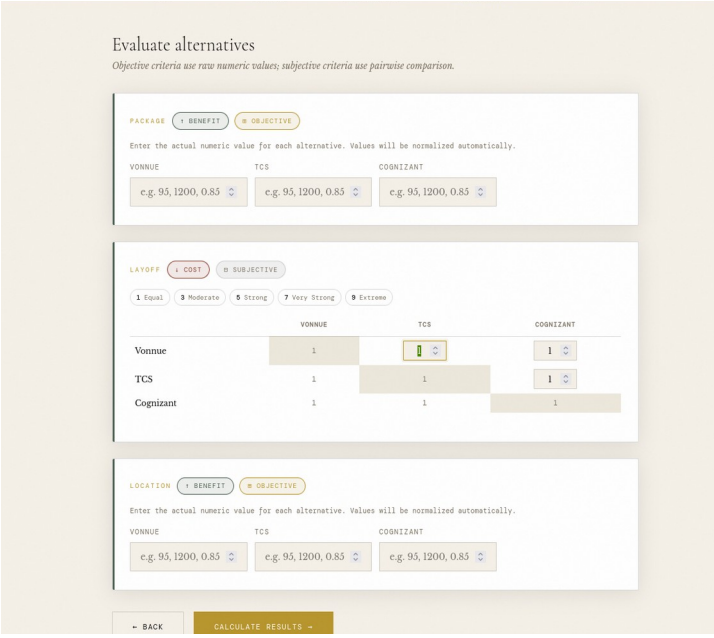
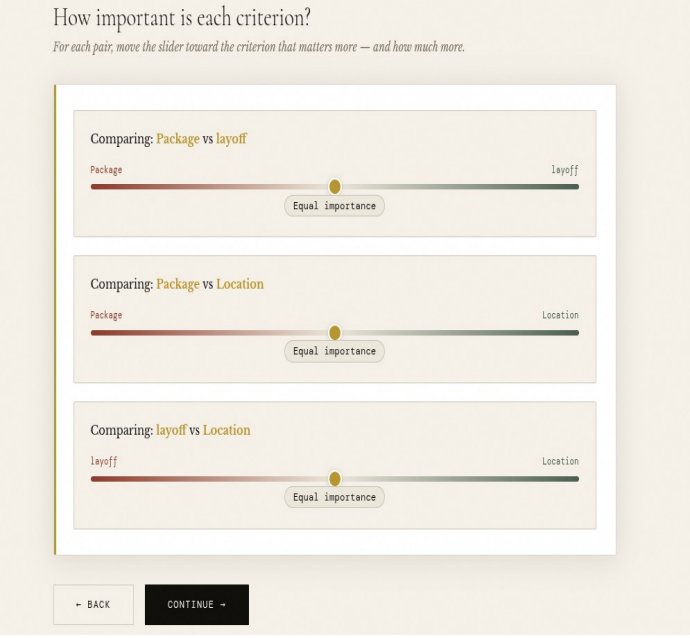
Formula:

Score=Return−λ×Variance

Why Use It:

- Very strong academically
- Impressive in evaluation
- Industry-standard concept

Also I studied about other algorithms like Expected Utility Theory, Expected Utility Theory, Monte Carlo Simulation , Fuzzy AHP.

Then I deployed the new website using vercel.

Since my old UI is not preffered for all users, I changed default option to objective and the subjective matrix part is converted to most user friendly interface according to the review by some friends.